



Comparing 2D and 3D p-Value Visualizations

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Introduction

Existing R packages such as `pvaluefunctions` and `concurve` visualize p-values in **two dimensions**. They show how the p-value changes with **one parameter** at a time — typically the effect size, given a fixed standard error.

The **p3d** package extends this to **three dimensions**, showing the p-value as a function of **both effect size and sample size** simultaneously.

This vignette compares the two approaches.

Setup

We will compare a 2D p-value curve (from `pvaluefunctions`) with a 3D p-value surface (from `p3d`), using the same scenario:

- Observed effect size: $d = 0.5$
- Sample size: $n = 30$

2D p-value curve



```
install.packages("pvaluefunctions")
```

Now we compute the 2D curve:

```
library(pvaluefunctions)

conf_dist(
  estimate = 0.5,
  stderr = 1 / sqrt(30),
  df = 29,
  tstat = 0.5 * sqrt(30),
  type = "ttest",
  plot_type = "p_val"
)
```

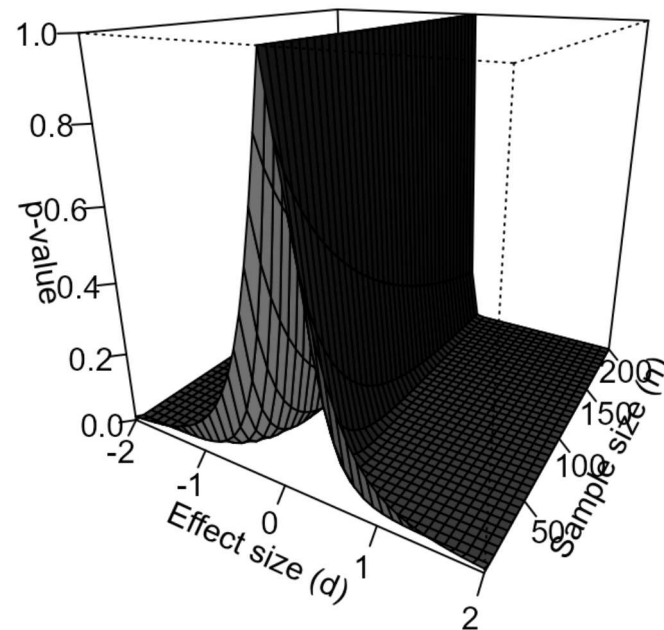
The resulting plot shows the p-value as a function of the estimated effect size, with the observed effect size (0.5) in the center. The curve is **symmetric** and has a **minimum** at the observed effect size.

3D p-value surface

The `p3d` package visualizes the p-value as a function of **both** effect size and sample size:

```
surface <- p_surface(
  d = seq(-2, 2, 0.1),
  n = seq(5, 200, 5)
)
```

p-Value Surface (static)



The 3D surface reveals structure that the 2D curve cannot:

1. **The p-value is not symmetric in n .** As n increases, the surface flattens toward zero for large $|d|$ and toward one near $d = 0$.
2. **The significance boundary is a curve, not a point.** For each n , there is a critical effect size $d_{\text{crit}}(n)$ above which the p-value falls below 0.05.

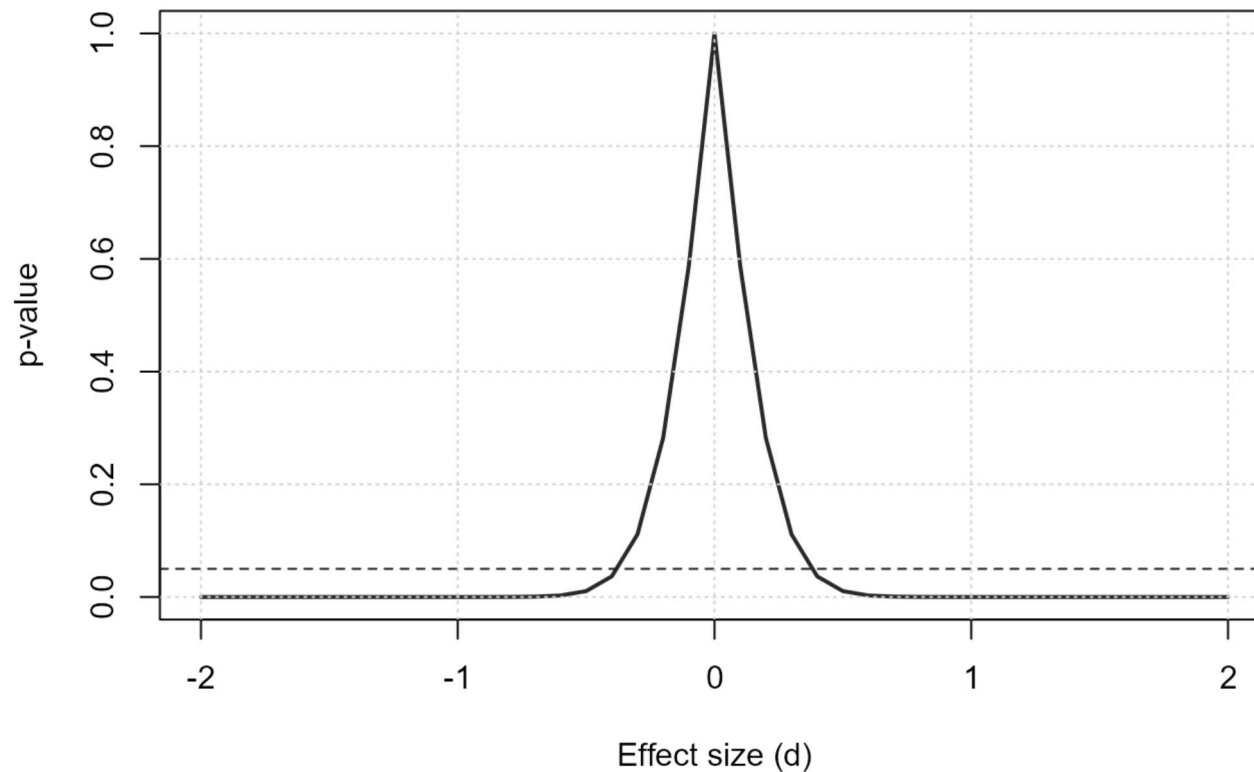


Comparing the two views

Let's place the 2D curve and the 3D surface side by side. The 2D curve corresponds to a **slice** of the 3D surface at $n = 30$:

```
# 2D curve at n = 30
curve_30 <- p_curve_2d(d = seq(-2, 2, 0.1), n = 30)

# Plot the 2D curve
plot(curve_30$d, curve_30$p, type = "l", lwd = 2, col = "darkred",
      xlab = "Effect size (d)", ylab = "p-value",
      main = "2D p-value curve (n = 30)")
abline(h = 0.05, col = "blue", lty = 2)
grid()
```



This 2D curve is mathematically equivalent to the **slice** of the 3D surface at $n = 30$. Both give the same p-value at $d = 0.5$:

```
curve_30$p[curve_30$d == 0.5]
#> [1] 0.01043739
surface$p[surface$d == 0.5, surface$n == 30]
#> [1] 0.01043739
```



three different sample sizes:

```
surface$p[surface$d == 0.5, surface$n == 10]
#> [1] 0.1483047
surface$p[surface$d == 0.5, surface$n == 30]
#> [1] 0.01043739
surface$p[surface$d == 0.5, surface$n == 100]
#> [1] 2.481396e-06
```

At $n = 10$, the p-value is **above** 0.05 (not significant). At $n = 30$, it is **below** 0.05 (significant). At $n = 100$, it is **far below** 0.05 (highly significant).

This is the moderating effect of sample size. The same effect size produces different conclusions depending on n . The 2D curve cannot show this — it is fixed at one n .

Jensen's inequality: two interpretations of the p-value

A subtle but important point: the p-value has **two different interpretations**, which the 3D surface helps clarify.

Conditional p-value (fixed d): the p-value computed for a *specific* effect size. This is what `p_surface()` returns:

$$p_{\text{conditional}}(d, n) = 2 \left[1 - F_{t(n-1)}(d\sqrt{n}) \right]$$

For $d = 0.5, n = 30$:

```
p_surface(d = 0.5, n = 30)$p[1, 1]
```



$$\mathbb{E}[p(d_{\text{obs}}, n)] = \int p(d_{\text{obs}}, n) \cdot f(d_{\text{obs}} \mid d, n) dd_{\text{obs}}$$

For $d = 0.5, n = 30$, this is approximately:

```
set.seed(42)
n_sim <- 10000
n <- 30
d <- 0.5

p_values <- replicate(n_sim, {
  x <- rnorm(n, mean = d, sd = 1)
  t.test(x, mu = 0)$p.value
})

mean(p_values)
#> [1] 0.05887669
```

The two values differ substantially (0.0104 vs 0.0589). This is due to **Jensen's inequality**: the p-value is a convex function of d , so the expected p-value is **larger** than the conditional p-value.

The `p3d` package visualizes the **conditional** p-value, consistent with power analysis and existing packages. The **expected** p-value is a subject of ongoing research.

Conclusion

The 2D and 3D visualizations of the p-value are complementary:

- **2D**: Ideal for examining a single scenario in detail



continuous nature of the significance boundary, and the interaction between effect size and sample size.

By making these features visible, **p3d** helps weaken the dichotomous thinking that reduces the p-value to a binary decision.

References

- Helske, J., Helske, S., & Cooper, M. (2021). *Visualizing Hypothesis Tests in Practice*. The R Journal.
- Infanger, D., & Schmidt-Trucksäss, A. (2019). *P value functions: An underused method to present research results and to promote quantitative reasoning*. Statistics in Medicine.